KAIZEN MATHS

Lesson starter

Number Bases and Number Sets

Number Sets

Name _____

S1

Classify -2.75. List all the number sets it belongs to.

S2

State the smallest number set that contains 0.

S3

True or false: Every irrational number is real.

S4

Classify 7. List all the number sets it belongs to.

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NUM-5141-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Number Bases and Number Sets

Number Sets

Name _____

E1

State the smallest number set that contains -6.

E2

True or false: Every integer is a whole number.

E3

Classify √2. List all the number sets it belongs to.

KAIZEN MATHS

Exit ticket

Number Bases and Number Sets

Number Sets

Name _____

E1

State the smallest number set that contains -6.

E2

True or false: Every integer is a whole number.

E3

Classify √2. List all the number sets it belongs to.

NUM-5141-L1 · E

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State the smallest number set that contains -6.

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True or false: Every integer is a whole number.

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NUM-5141-L1 · E

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NUM-5141-L1 · E

Show your working

Teacher answers & working

Number Bases and Number Sets · Number Sets

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S1

Classify -2.75. List all the number sets it belongs to.

Answer:

$\mathbb{Q}$, $\mathbb{R}$

Working:

Check what kind of number -2.75 is.

-2.75 is a terminating decimal, so it can be written as a fraction.

Therefore -2.75 belongs to: $\mathbb{Q}$, $\mathbb{R}$.

S2

State the smallest number set that contains 0.

Answer:

$\mathbb{W}$

Working:

Start with the most specific description of 0.

0 is a whole number but not a positive counting number under this convention.

The smallest suitable set is W: whole number.

S3

True or false: Every irrational number is real.

Answer:

True

Working:

Read the statement carefully.

The real numbers contain both rational and irrational numbers.

Therefore the statement is true.

S4

Classify 7. List all the number sets it belongs to.

Answer:

$\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Working:

Check what kind of number 7 is.

7 is a positive counting number.

Therefore 7 belongs to: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.

Teacher answers & working

Number Bases and Number Sets · Number Sets

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E1

State the smallest number set that contains -6.

Answer:

$\mathbb{Z}$

Working:

Start with the most specific description of -6.

-6 is an integer, so it is also rational and real.

The smallest suitable set is $\mathbb{Z}$: integer.

E2

True or false: Every integer is a whole number.

Answer:

False

Working:

Read the statement carefully.

Negative integers such as -3 are not whole numbers.

Therefore the statement is false.

E3

Classify √2. List all the number sets it belongs to.

Answer:

irrational, $\mathbb{R}$

Working:

Check what kind of number √2 is.

√2 cannot be written as a fraction of integers.

Therefore √2 belongs to: irrational, $\mathbb{R}$.